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Lecture 28

Momentum space RG for gaussian model

The generalized hamiltonian is given by

$$\mathcal{K}(\sigma) = -\frac{1}{2} \int_{\vec{q}} (\tau + q^2) \sigma_{\vec{q}} \sigma_{-\vec{q}}$$

The partition function is given by

$$\mathcal{Z} = \int_{\sigma_{\vec{q}}} e^{\mathcal{K}(\sigma_{\vec{q}})}$$

Here ' σ ' is a free variable which is present at every ' $\vec{q}$ ' & can take values b/w $-\infty$ & $+\infty$.

Integrating out $|\vec{q}| > 1/2$ implies that we integrate over all $\sigma_{\vec{q}}$ whose $|\vec{q}| > 1/2$. Thus for ease of notations we denote,

$$\sigma_{\vec{q}} = \begin{cases} \sigma_{\vec{q}}^{(0)} & , 0 < |\vec{q}| < \frac{1}{2} \\ \sigma_{\vec{q}}^{(1)} & , \frac{1}{2} < |\vec{q}| < 1 \end{cases}$$

$$Z = \int_{\vec{\sigma}_{\vec{r}}} e^{\mathcal{K}(\vec{\sigma}_{\vec{r}})} = \int_{\vec{r}} \prod_{\vec{r}} d\vec{\sigma}_{\vec{r}} e^{\mathcal{K}(\vec{\sigma}_{\vec{r}})}$$

$$= \underbrace{\int_{\vec{r}} \prod_{\vec{r}} d\vec{\sigma}_{\vec{r}}^{(0)}}_{0 \leq r < \frac{1}{2}} \underbrace{\int_{\vec{r}} \prod_{\vec{r}} d\vec{\sigma}_{\vec{r}}^{(1)}}_{\frac{1}{2} \leq r < 1} \exp \left[-\frac{1}{2} \int_{r=0}^{r=1} (\partial + r^2) \vec{\sigma}_{\vec{r}} \cdot \vec{\sigma}_{-\vec{r}} \right]$$

$$e^{\mathcal{K}(\vec{\sigma}_{\vec{r}})} = \exp \left[-\frac{1}{2} \int_{r=0}^{r=1} (\partial + r^2) \vec{\sigma}_{\vec{r}} \cdot \vec{\sigma}_{-\vec{r}} \right]$$

$$= \exp \left[\left(-\frac{1}{2} \int_{r=0}^{r=\frac{1}{2}} (\partial + r^2) \vec{\sigma}_{\vec{r}}^{(0)} \cdot \vec{\sigma}_{-\vec{r}}^{(0)} \right) + \left(-\frac{1}{2} \int_{r=\frac{1}{2}}^{r=1} (\partial + r^2) \vec{\sigma}_{\vec{r}}^{(1)} \cdot \vec{\sigma}_{-\vec{r}}^{(1)} \right) \right]$$

$$= \exp \left[-\frac{1}{2} \int_{r=0}^{r=\frac{1}{2}} (\partial + r^2) \vec{\sigma}_{\vec{r}}^{(0)} \cdot \vec{\sigma}_{-\vec{r}}^{(0)} \right] \exp \left[-\frac{1}{2} \int_{r=\frac{1}{2}}^{r=1} (\partial + r^2) \vec{\sigma}_{\vec{r}}^{(1)} \cdot \vec{\sigma}_{-\vec{r}}^{(1)} \right]$$

$$\Rightarrow Z = \int_{\vec{r}} \prod_{\vec{r}} d\vec{\sigma}_{\vec{r}}^{(0)} \exp \left[-\frac{1}{2} \int_{r=0}^{r=\frac{1}{2}} (\partial + r^2) \vec{\sigma}_{\vec{r}}^{(0)} \cdot \vec{\sigma}_{-\vec{r}}^{(0)} \right]$$

$$\times \int_{\vec{r}} \prod_{\vec{r}} d\vec{\sigma}_{\vec{r}}^{(1)} \exp \left[-\frac{1}{2} \int_{r=\frac{1}{2}}^{r=1} (\partial + r^2) \vec{\sigma}_{\vec{r}}^{(1)} \cdot \vec{\sigma}_{-\vec{r}}^{(1)} \right]$$

$$\Rightarrow Z = \int_{\sigma_{\vec{q}}^{(0)}} e^{\mathcal{H}(\sigma_{\vec{q}}^{(0)})} \int_{\sigma_{\vec{q}}^{(1)}} e^{\mathcal{H}(\sigma_{\vec{q}}^{(1)})}$$

Thus the partition function is easily separable in the Gaussian model. Now we just have to integrate out the second functional integral over $\sigma_{\vec{q}}^{(1)}$.

$$\int_{\sigma_{\vec{q}}^{(1)}} e^{\mathcal{H}(\sigma_{\vec{q}}^{(1)})} = \int_{\sigma_{\vec{q}}^{(1)}} \exp \left[\underbrace{-\frac{1}{2} \sum_{q=\frac{1}{2}}^{q=1} (\partial + q^2) \sigma_{\vec{q}}^{(1)} \sigma_{-\vec{q}}^{(1)}}_{\text{convert the integral to sum}} \right]$$

$$= \int \left[\prod_{q=\frac{1}{2}}^{q=1} d\sigma_{\vec{q}}^{(1)} \right] \exp \left[-\frac{1}{2N} \sum_{q=\frac{1}{2}}^{q=1} (\partial + q^2) \sigma_{\vec{q}}^{(1)} \sigma_{-\vec{q}}^{(1)} \right]$$

$$= \int \left[\prod_{q=\frac{1}{2}}^{q=1} d\sigma_{\vec{q}}^{(1)} \right] \prod_{q=\frac{1}{2}}^{q=1} \exp \left[-\frac{1}{2N} (\partial + q^2) \sigma_{\vec{q}}^{(1)} \sigma_{-\vec{q}}^{(1)} \right]$$

$$\text{use } \prod_{q=\frac{1}{2}}^{q=1} d\sigma_{\vec{q}}^{(1)} = \prod_{q=\frac{1}{2}}^{q=1} 2 d\sigma_{\vec{q}}^{(R)} d\sigma_{\vec{q}}^{(I)} \quad (\text{as seen earlier})$$

& since, $\sigma_{\vec{q}} = \sigma_{\vec{q}}^{(R)} + i\sigma_{\vec{q}}^{(I)}$ & $\sigma_{-\vec{q}} = \sigma_{\vec{q}}^{(R)} - i\sigma_{\vec{q}}^{(I)}$

$$\Rightarrow \sigma_{\vec{q}}^{(1)} \sigma_{-\vec{q}}^{(1)} = \sigma_{\vec{q}}^{(R)^2} + \sigma_{\vec{q}}^{(I)^2}$$

where $\sigma_{\vec{q}}^{(R)}$ & $\sigma_{\vec{q}}^{(I)}$ varies b/w $-\infty$ & $+\infty$

Thus we can write $\int_{\sigma_{\vec{q}}^{(1)}} e^{\mathcal{K}(\sigma_{\vec{q}}^{(1)})}$ as

$$= \prod_{q=\frac{1}{2}}^{q=1} 2 \int d\sigma_{\vec{q}}^{(R)} d\sigma_{\vec{q}}^{(I)} \exp \left[-\frac{(\gamma + q_v^2)}{2N} \left(\sigma_{\vec{q}}^{(R)^2} + \sigma_{\vec{q}}^{(I)^2} \right) \right]$$

$$= \prod_{q=\frac{1}{2}}^{q=1} 2 \int d\sigma_{\vec{q}}^{(R)} \exp \left[-\frac{(\gamma + q_v^2)}{2N} \sigma_{\vec{q}}^{(R)^2} \right] \\ \times \int d\sigma_{\vec{q}}^{(I)} \exp \left[-\frac{(\gamma + q_v^2)}{2N} \sigma_{\vec{q}}^{(I)^2} \right]$$

using,

$$\int_{-\infty}^{\infty} d\sigma_{\vec{q}}^{(R)} \exp \left[-\frac{(\gamma + q_v^2)}{2N} \sigma_{\vec{q}}^{(R)^2} \right] = \sqrt{\frac{2N\pi}{(\gamma + q_v^2)}}$$

$$\& \int_{-\infty}^{\infty} d\sigma_{\vec{q}}^{(I)} \exp \left[-\frac{(\gamma + q_v^2)}{2N} \sigma_{\vec{q}}^{(I)^2} \right] = \sqrt{\frac{2N\pi}{(\gamma + q_v^2)}}$$

$$\begin{aligned}\Rightarrow Z &= \int_{\sigma_{\vec{q}}^{(0)}} e^{\mathcal{H}(\sigma_{\vec{q}}^{(0)})} \int_{\sigma_{\vec{q}}^{(1)}} e^{\mathcal{H}(\sigma_{\vec{q}}^{(1)})} \\ &= \int_{\sigma_{\vec{q}}^{(0)}} e^{\mathcal{H}(\sigma_{\vec{q}}^{(0)})} \prod_{q=\frac{1}{2}}^{q=1} 2 \left[\frac{2N\pi}{(\gamma + q^2)} \right]\end{aligned}$$

$$\Rightarrow Z = e^{\tilde{\mathcal{A}}} \int_{\sigma_{\vec{q}}^{(0)}} e^{\mathcal{H}(\sigma_{\vec{q}}^{(0)})}$$

where $e^{\tilde{\mathcal{A}}} = \prod_{q=\frac{1}{2}}^{q=1} 2 \left[\frac{2N\pi}{(\gamma + q^2)} \right]$

$$\Rightarrow \tilde{\mathcal{A}} = \sum_{q=\frac{1}{2}}^{q=1} \log \left[\frac{4N\pi}{(\gamma + q^2)} \right] = \text{Constant}$$

$$\Rightarrow Z = e^{\tilde{\mathcal{A}}} \int_{\sigma_{\vec{q}}^{(0)}} \exp \left[-\frac{1}{2} \int_{q=0}^{q=\frac{1}{2}} (\gamma + q^2) \sigma_{\vec{q}}^{(0)} \sigma_{-\vec{q}}^{(0)} \right]$$

Thus the coarse graining procedure is done & we are left with rescaling.

Define $\vec{q} \rightarrow \vec{q}' = 2\vec{q} \rightarrow \vec{q} = \frac{\vec{q}'}{2}$
 Then the integration limit for $|\vec{q}'|$ goes from 0 to 1.

$$\begin{aligned} \Rightarrow \mathcal{K} &= \int_{q=0}^{q=\frac{1}{2}} (d^d q) (\alpha + q^2) \sigma_{\vec{q}}^{(0)} \sigma_{-\vec{q}}^{(0)} \\ &= -\frac{1}{2} \int_{q'=0}^{q'=1} 2^{-d} (d^d q') \left(\alpha + \frac{q'^2}{4} \right) \sigma_{\frac{\vec{q}'}{2}}^{(0)} \sigma_{-\frac{\vec{q}'}{2}}^{(0)} \\ &= -\frac{1}{2} \int_{q'=0}^{q'=1} (d^d q') \left[2^{-d} \alpha \sigma_{\frac{\vec{q}'}{2}}^{(0)} \sigma_{-\frac{\vec{q}'}{2}}^{(0)} + \frac{2^{-d}}{4} q'^2 \sigma_{\frac{\vec{q}'}{2}}^{(0)} \sigma_{-\frac{\vec{q}'}{2}}^{(0)} \right] \end{aligned}$$

Redefine σ 's such that the coefficient of $\sigma_{\frac{\vec{q}'}{2}}^{(0)} \sigma_{-\frac{\vec{q}'}{2}}^{(0)} q'^2 = 1$. To resemble the old $\mathcal{K}$.

$$\Rightarrow \sigma_{\frac{\vec{q}'}{2}}^{(0)} = \zeta \sigma_{\vec{q}'} \quad \& \quad \sigma_{-\frac{\vec{q}'}{2}}^{(0)} = \zeta \sigma_{-\vec{q}'}$$

$$\text{where } \frac{2^{-d}}{4} q'^2 \sigma_{\frac{\vec{q}'}{2}}^{(0)} \sigma_{-\frac{\vec{q}'}{2}}^{(0)} = \underbrace{\frac{2^{-d}}{4} \zeta^2}_{1} q'^2 \sigma_{\vec{q}'} \sigma_{-\vec{q}'}$$

$$\Rightarrow \underline{\zeta^2 = 4(2^d)} \Rightarrow \sigma_{\frac{\vec{q}'}{2}}^{(0)} \sigma_{-\frac{\vec{q}'}{2}}^{(0)} = 4(2^d) \sigma_{\vec{q}'} \sigma_{-\vec{q}'}$$

Thus the rescaled hamiltonian is

$$\mathcal{H}' = -\frac{1}{2} \int_{q'=0}^{q'=1} (\delta' + q'^2) \sigma_{\vec{q}'}^+ \sigma_{-\vec{q}'}^+$$

where $\delta' = 4\delta$ & the new partition function is,

$$\mathcal{Z}' = e^{\tilde{\mathcal{A}}} \int \left[\prod_{q'=0}^{q'=1} \mathcal{L}_3 d\sigma_{\vec{q}'}^+ \right] e^{\mathcal{H}'(\sigma_{\vec{q}'}^+)}$$

$$\mathcal{Z}' = e^{\tilde{\mathcal{A}}} \left[\prod_{q'=0}^{q'=1} \mathcal{L}_3 \right] \int \prod_{q'=0}^{q'=1} d\sigma_{\vec{q}'}^+ e^{\mathcal{H}'(\sigma_{\vec{q}'}^+)}$$

$$= e^{\tilde{\mathcal{A}}} e^{\sum_{\vec{q}'} \log(\mathcal{L}_3)} \int \prod_{q'=0}^{q'=1} d\sigma_{\vec{q}'}^+ e^{\mathcal{H}'(\sigma_{\vec{q}'}^+)}$$

$$\text{Let } \mathcal{A} = \tilde{\mathcal{A}} + \sum_{\vec{q}'} \log(\mathcal{L}_3) \quad \& \quad \int \prod_{q'=0}^{q'=1} d\sigma_{\vec{q}'}^+ = \int_{\sigma_{\vec{q}'}^+}$$

Thus the final rescaled partition function is,

$$\mathcal{Z}' = e^{\mathcal{A}} \int_{\sigma_{\vec{q}'}^+} e^{\mathcal{H}'(\sigma_{\vec{q}'}^+)}$$

Scaling form of Correlation function

The inverse fourier transform of the $\sigma_{\vec{q}}$ in the continuous $|\vec{q}|$ limit where $0 \leq |\vec{q}| \leq 1$ is the spin field $\mathcal{S}(\vec{R})$ defined as

$$\mathcal{S}(\vec{R}) = \frac{1}{(2\pi)^d} \int_{q=0}^{q=1} (d^d q) \sigma_{\vec{q}} e^{i\vec{q} \cdot \vec{R}}$$

Then the real space correlation function is $\Gamma(\vec{R}) = \langle \mathcal{S}(0) \mathcal{S}(\vec{R}) \rangle$

$$\rightarrow \Gamma(\vec{R}) = \frac{1}{Z} \int_{\sigma_{\vec{q}}} \mathcal{S}(0) \mathcal{S}(\vec{R}) e^{K(\mathcal{S})}$$

Applying fourier transform on ' $\mathcal{S}$ ' we get,

$$\mathcal{S}(0) \mathcal{S}(\vec{R}) = \int_{\vec{q}} \int_{\vec{q}'} \sigma_{\vec{q}} \sigma_{\vec{q}'} e^{i\vec{q} \cdot \vec{R}}$$

$$\& K(\mathcal{S}) \rightarrow \mathcal{H}(\sigma_{\vec{q}})$$

$$\Rightarrow \Gamma(\vec{R}) = \frac{1}{Z} \int_{\vec{\sigma}} \int_{\vec{q}} \int_{\vec{q}'} \sigma_{\vec{q}} \sigma_{\vec{q}'} e^{i\vec{q} \cdot \vec{R}} e^{\mathcal{H}(\sigma_{\vec{q}})}$$

$$\Rightarrow \Gamma(\vec{R}) = \int_{\vec{q}} e^{i\vec{q} \cdot \vec{R}} \left[\frac{1}{Z} \int_{\vec{\sigma}} \sigma_{\vec{q}} \sigma_{\vec{q}'} e^{\mathcal{H}(\sigma_{\vec{q}})} \right]$$

If we denote $\langle \sigma_{\vec{q}} \sigma_{\vec{q}'} \rangle$ as the correlation b/w spin variables in momentum space, then,

$$\langle \sigma_{\vec{q}} \sigma_{\vec{q}'} \rangle = \frac{1}{Z} \int_{\vec{\sigma}} \sigma_{\vec{q}} \sigma_{\vec{q}'} e^{\mathcal{H}(\sigma_{\vec{q}})}$$

$$\& \Gamma(\vec{R}) = \int_{\vec{q}} e^{i\vec{q} \cdot \vec{R}} \int_{\vec{q}'} \langle \sigma_{\vec{q}} \sigma_{\vec{q}'} \rangle$$

One can show that $\langle \sigma_{\vec{q}} \sigma_{\vec{q}'} \rangle$ is a functional integral that turns out to be proportional to $\delta(\vec{q} + \vec{q}')$ by demanding ' $\mathcal{S}$ ' is a real field.

$$\Rightarrow \langle \sigma_{\vec{q}_1} \sigma_{\vec{q}_2} \rangle = (2\pi)^d \delta(\vec{q}_1 + \vec{q}_2) \Gamma(\vec{q}_1, \gamma)$$

$[\gamma \propto (T - T_c)]$

If $\vec{q}_1$ & $\vec{q}_2$ are such that they lie inside the inner sphere which implies $0 < |\vec{q}_1| < 1/2$ & $0 < |\vec{q}_2| < 1/2$, then in one step of RG procedure,

$$\begin{aligned} \langle \sigma_{\vec{q}_1} \sigma_{\vec{q}_2} \rangle &= \frac{1}{Z} \int_{\sigma_{\vec{q}}} \sigma_{\vec{q}_1} \sigma_{\vec{q}_2} e^{\mathcal{H}(\sigma_{\vec{q}})} \\ &= \frac{1}{Z'} \int_{\sigma_{\vec{q}'}}^2 \sigma_{\vec{q}'_1} \sigma_{\vec{q}'_2} e^{\mathcal{H}'(\sigma_{\vec{q}'})} \end{aligned}$$

where $\vec{q}'_1 = 2\vec{q}_1$, $\vec{q}'_2 = 2\vec{q}_2$

& $\mathcal{H}'(\sigma_{\vec{q}'}) = -\frac{1}{2} \int_{q'=0}^{q'=1} (\gamma' + q'^2) \sigma_{\vec{q}'} \sigma_{-\vec{q}'}$

where $\gamma' = 4\gamma$

$$\Rightarrow \langle \sigma_{\vec{q}_1} \sigma_{\vec{q}_2} \rangle = (2\pi)^d \delta(\vec{q}_1 + \vec{q}_2) \Gamma(\vec{q}_1, \chi)$$

↓ 1 RG procedure

$$\langle \sigma_{\vec{q}_1} \sigma_{\vec{q}_2} \rangle = (2\pi)^d \mathcal{C}^2 \delta(2\vec{q}_1 + 2\vec{q}_2) \Gamma(2\vec{q}_1, \chi')$$

$$\Rightarrow (2\pi)^d \delta(\vec{q}_1 + \vec{q}_2) \Gamma(\vec{q}_1, \chi) = (2\pi)^d \mathcal{C}^2 \delta(2\vec{q}_1 + 2\vec{q}_2) \Gamma(2\vec{q}_1, \chi')$$

$$\begin{aligned} \Rightarrow \delta(\vec{q}_1 + \vec{q}_2) \Gamma(\vec{q}_1, \chi) &= \mathcal{C}^2 \delta(2\vec{q}_1 + 2\vec{q}_2) \Gamma(2\vec{q}_1, \chi') \\ &= \underbrace{\mathcal{C}^2 (2^{-d})}_4 \delta(\vec{q}_1 + \vec{q}_2) \Gamma(2\vec{q}_1, 4\chi) \end{aligned}$$

$$\Rightarrow \underline{\Gamma(\vec{q}_1, \chi) = 4 \Gamma(2\vec{q}_1, 4\chi)}$$

Applying inverse fourier transform on the above eqⁿ we get,

$$\begin{aligned} (2\pi)^{-d} \int_{\vec{q}_1} (d^d q_1) \Gamma(\vec{q}_1, \chi) e^{i\vec{q}_1 \cdot \vec{R}} \\ = (2\pi)^{-d} \int_{\vec{q}_1} (d^d q_1) [4 \Gamma(2\vec{q}_1, 4\chi)] e^{i\vec{q}_1 \cdot \vec{R}} \end{aligned}$$

$$\begin{aligned}
\Gamma(\vec{R}, \chi) &= (2\pi)^{-d} \int_{q_1} (d^d q_1) [4 \Gamma(2\vec{q}_1, 4\chi)] e^{i\vec{q}_1 \cdot \vec{R}} \\
&= (2\pi)^{-d} \int_{q_1} \frac{d^d(2q_1)}{2^d(2^{-2})} \Gamma(2\vec{q}_1, 4\chi) e^{i(2\vec{q}_1 \cdot \frac{\vec{R}}{2})} \\
&= \frac{1}{2^{d-2}} (2\pi)^{-d} \int_{q'_1} (d^d q'_1) \Gamma(q'_1, \chi) e^{i\vec{q}'_1 \cdot \frac{\vec{R}}{2}} \\
&= 2^{2-d} \Gamma\left[\frac{\vec{R}}{2}, 4\chi\right] \\
\Rightarrow \Gamma(\vec{R}, \chi) &\xrightarrow[\text{once}]{\text{R.G.}} 2^{2-d} \Gamma\left[\frac{\vec{R}}{2}, 4\chi\right]
\end{aligned}$$

After l -applications of RG we get

$$\Gamma(\vec{R}, \chi) = (2^{2-d})^l \Gamma\left[\frac{\vec{R}}{2^l}, 2^{2l}\chi\right]$$

Since Correlation function depends only on magnitude of distance, thus we can write

$$(R = |\vec{R}|) \quad \Gamma(R, \chi) = (2^{2-d})^l \Gamma\left[\frac{R}{2^l}, 2^{2l}\chi\right]$$

Choose 'l' such that $\frac{R}{2^l} = 1 \Rightarrow 2^l = R^1$

$$\begin{aligned} \Rightarrow \Gamma(R, \gamma) &= (R)^{2-d} \Gamma(1, R^2 \gamma) \\ &= \frac{1}{R^{d-2}} \Gamma(1, [R \sqrt{\gamma}]^2) \\ &= \frac{1}{R^{d-2}} \Gamma\left(1, \left[\frac{R}{\gamma^{-1/2}}\right]^2\right) \end{aligned}$$

Some function of $\frac{R}{\gamma^{-1/2}}$

$$\Rightarrow \Gamma(R, \gamma) = \frac{1}{R^{d-2}} \Upsilon\left(\frac{R}{\gamma^{-1/2}}\right)$$

$\gamma \sim |t| \rightarrow$ reduced temperature
Reciprocal of Coefficient of R is ' ξ_g '.

$$\rightarrow \xi_g = \gamma^{-1/2} \sim |t|^{-1/2}$$

$$\rightarrow \xi_g = \text{Correlation length} \sim |t|^{-1/2}$$

$$G = \frac{\Upsilon(R/\xi_g)}{\gamma^{d-2+\eta}} \Rightarrow \eta = 0 \text{ \& } \xi_g \sim |t|^{-\nu} \Rightarrow \nu = \frac{1}{2}$$

} defⁿ of η

Thus Gaussian model reproduces the mean field critical exponents.